3.6 EXPLANATION IN NON-PHYSICAL SCIENCES

There is a view that explanation in biology, psychology, and the social sciences has the same structure as in the physical sciences. But the causal type of explanation is essentially inadequate in fields other than physics and chemistry. And it is more so in the case of social sciences which involve explanation of purposive behaviour.

Hempel and Oppenheim point out to some of these limitation of applying models of scientific explanation to social sciences. First, events involving the activities of humans singly or in groups have a peculiar uniqueness and irrepeatability which make them inaccessible to causal explanation. Causal explanation presupposes uniformities and repeatability which may not hold for phenomena in social sciences.

Second, the establishment of scientific generalizations and explanatory principles for human behaviour is impossible because the reactions of an individual in a given situation depend not only upon that situation, but upon the previous history of the individual.

Third, the explanation of any phenomenon involving purposive behaviour calls for reference to motivations and thus for teleological rather than causal analysis. Many of the explanations which are offered for human actions involve reference to goals and motives. Motivational explanations often tend to have less cognitive significance.

Attempts to extend scientific models of explanation to non-physical sciences without closer understanding of the context often lead to inappropriate explanations. One possibility is that there are contexts or areas within social sciences like economics where there is room for application of scientific or to be specific, positivist models of explanations. But at the same time there are vast aspects of economics where such application may not be easily amenable.

Check Your Progress 3

1) State the difference between laws and ordinary statement.

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2) Describe two limitations of application of scientific models to social sciences.

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3.7 LET US SUM UP

Positivism as an approach to the pursuit of knowledge is procrustean in nature. It has passed through several stages from radical empiricism to logical positivism to logical empiricism. Its propositions, often contested, evolved over a long period. The main characteristics (unified view) of positivism lie in a set of its four rules-phenomenalism, nouminalism, the rule of value free statements and the rule of unity of science.

Testability under the positivism has been basic criteria for distinguishing scientific knowledge from non-scientific or meaningless statement. In early phase, emphasis was on verification. Later on Karl Popper suggested falsification testing in place of verification. Subsequently, emphasis shifted from falsification to confirmation.

Rules of logic have become important part of the whole structure of scientific explanation under positivism. There are two types of deductive reasoning-categorical syllogism and hypothetical syllogism.

Theories are the basic building blocks of scientific explanation. Hence the status, structure and functions of theories and theoretical terms are essential parts of scientific explanation.

The basic modes of scientific explanation within the positivist approach are: Hypothetico-Deductive Model (HD), Deductive-Nomological (DN) Model, and Inductive Probabilistic (IP) models. The DN and IP models are subject to several limitations. The application of scientific models to social sciences has several limitations; peculiar uniqueness of human activities, difficulty in scientific generalizations and explanatory principles for human behaviour, etc. Hence applications of scientific models to non-physical sciences without closer understanding of the context often lead to inappropriate explanation.

3.8 KEY WORDS

Deduction : Deduction is a process used to derive particular statements from general statements.

Empiricism : 'Empiricism' refers to distinctive approach that envisages that all knowledge comes from the senses.

Falsification : The concept developed by Karl Popper (1959) is a philosophy of Science also known as critical rationalism. It uses the logic of deduction to provide the foundation for the Hypothetico-Deductive Method. He rejected the idea that observation provides the formulation for scientific theories. Theories are invented to account for observation, not derived from them. Observations are used to try to reject false theories. Theories that survive this critical process are provisionally accepted but never proven to be true.

Induction : Induction is a logical process of moving from particular statements to general statements. This is used in the social sciences to produce theory from data.

Logical Positivism : An approach to philosophy of science that envisages that all meaningful statements are either empirical or logical in character. This means that they are either open to test by observational evidence or are matches of definition and therefore, tautological. According to this approach, no fundamental difference exists between natural and social sciences.

Modelling : Refers to a process of creating a simplified representation of a System or phenomena with hypotheses to describe the system or explain the phenomenon.

Theory : Refers to set of general laws that are related in networks and are generated by inductive or deductive form of logic. In other words, theory can be described as a set of propositions of different levels of generality.

Verification : Refers to the use of empirical data, observation, test or experiment to confirm the truth or rational justification of a hypothesis.

3.9 SOME USEFUL BOOKS

Brody, Baruch A. (1970); *Readings in the Philosophy of Science*, Prentice Hall Inc., Englewoodcliffs, New Jersey.

Bryant, C.G.A (1985); *Positivism in Social Theory and Research,* London, Macmillan.

Caldwell, Bruce J (1984); *Beyond Positivism: Economic Methodology in the Twentieth Century,* George Allen and Unwin, Boston.

Hausman, D.M (1994); *The Philosophy of Economics: An Anthology,* Second Ed., Cambridge University Press, Cambridge.

Popper, Karl (1959); *The Logic of Scientific Discovery,* Basic Books, New York.

Stewart, F. (1979); *Reasoning and Method in Economics,* McGraw-Hill Book Co., London.

3.10 ANSWERS OR HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

1) Scientific explanation refers to give causes of phenomenon to be explained.

2) Phenomenon refers to a situation in which facts and objective facts based on experience forms the basis of knowledge. It is existence, not essence. On the other hand, nouminalism is insight representative of facts. It envisages that any abstract concepts used in scientific explanation must also be derived from experience.

3) Broadly there are two rules of logic. Categorical syllogism, and hypothetical syllogism.Logical formulation of two categorical statements serving as premises is categorical syllogism.

4) See Sub-section 3.4.4.

Check Your Progress 2

1) Facts, testing and formal logic.

2) See Sub-section 3.5.1

3) See Sub-section 3.5.2.1

Check Your Progress 3

1) Laws refer to universal statements expressing necessary truths whereas ordinary statement refers to particular statements of mere facts and express 'contingent truths'.

2) See Section 3.6

UNIT 4 DEBATES ON MODELS OF EXPLANATION IN ECONOMICS

Structure

4.0 OBJECTIVES

The unit aims at tracing the broad contours of the debate relating to bringing explanation in Economics closer to the Models of Positive Science. Patterning explanation in Economics on the lines of Positivist Models of Explanation has been a major methodological obsession in Economics. But the subject matter of Economics is not as easily amenable as physical science for application of rules of Positivism. After going through this unit, you will be able to:

- state the efforts made by mainstream economists to adopt the model of explanation of Positivism;

- explain the limitations, difficulties and differences in adopting 'scientific' models of explanation; and
- appraise the present state of explanation in Economics.

4.1 INTRODUCTION

The urge to model Economics or Political Economy as a science on the lines of physical sciences dates back to the period of Classical Economics. David Ricardo claimed that the principles of Political Economy were as accurate as the laws of gravity. Though Ricardo did not write much on the methods or models of explanation, his theories of political economy reflected **abstract-deductive thinking** which was claimed later as constituting basic methodology of Political Economy. It is 'Ricardo's habit of thinking' equated with hypothetico-deductive model of explanation that set the tone for the classical writers on methodology to carry forward the view of Political Economy as a pure science. We shall begin by briefly examining the views on the nature of Political Economy and its Method of Explanation in the Ricardian tradition as enunciated in the writings of N.W.Senior, J.S.Mill, J.E.Cairnes and J.N.Keynes. Then, we shall turn to Lionel Robbins who asserted that Economics was a pure positive science and emphasized apriorism as its method of explanation. This will be followed by T. Hutchison's insistence on empiricism if Economics were to be called science at all. The next two sections will discuss the foundations for present day mainstream practice of Economics in the form of 'instrumentalism' and 'operationalism' in the writings of Milton Friedman and Paul Samuelson respectively. Throughout, the focus is on the limitations of these approaches and the ad-hoc means by which problems are sought to be overcome. Though the unit is confined to the explanation in Economics in the Positivist mainstream, towards the end, a brief discussion of the alternative methods of explanation will be presented with a view to expose the student to the potential of plurality of methods of explanation in Economics.

4.2 CLASSICAL POLITICAL ECONOMY AND RICARDO'S METHOD

4.2.1 Ricardo's Method

The right method of economic enquiry as such was not discussed by Adam Smith. Smith's approach to methodology was, as it was analysed by later economists, somewhat *enigmatic*. There are instances which make some to point out that he was the first to raise political economy to the state of deductive science. But in the same breath there are many instances in his writings to as the founder of inductive historical method. Yet his overall approach often tends to be moral and ethical, distancing him from the attributes of any positivist explanation. For these reasons, on issues of methodology of Classical political economy, one invariably turns to David Ricardo.

Ricardo did not write on methodology of economics but his method explaining the principles of political economy is clearly discerned as positive, abstract and deductive, and it has come to be called '**hypothetico-deductive model of explanation'**. It vigorously denies that facts can speak for themselves and contends that it is theory that can make sense. Ricardo died in 1823, leaving behind the claim that political economy was a science. His **'Principles of Political**

Economy', which was a set of laws based on deductive reasoning, measures up to that claim. Questions were raised soon whether a deductive body of laws without empirical content could claim the status of science. The task of defending Ricardo's method and the claim of political economy as a science was left to his followers of classical political economy. The defence came from the writings of N.W.Senior, J.S.Mill, J.E.Cairnes and J.N.Keynes.

4.2.2 N.W. Senior

It was N.W.Senior in his 1927 lecture, which was later elaborated as **outline of the Science of Political Economy,** (1936), that first made a statement on the distinction between a pure and strictly positive science, and an impure and inherently normative *art* of economics. He went on to claim scientific status for the Ricardian system. Science of Political Economy, according to him, rests essentially on "a very few general propositions, which are the result of observation, or consciousness, and which almost every man, as soon as he hears them, admits, as being familiar to his thoughts". From this conclusions are drawn which hold true only in the absence of "particular disturbing causes".

Senior reduced these "very few general propositions" of political economy into the following four:

1) that every person desires to **maximize wealth with as little sacrifice as possible**;
2) that population tends to increase faster than the means of subsistence;
3) that labour working with machines is capable of producing a positive net product; and
4) that agriculture is subject to diminishing returns.

4.2.3 J.S. Mill

It is senior, who for the first time introduces the concept of '**economic man**' or 'maximizing man', as could be seen from the first proposition above. J.S.Mill, though staunch advocate of inductive method and a stout defender of methodological monism in science, was nonetheless sympathetic to Ricardo's deductive method of political economy. Mill, in his essay *on the Definition of Political Economy and on the Method of Investigation Proper* to It (1936), treats political economy as an "inexact science: and a 'separate science:, and thereby defends deductive method in economics. Elaborating on Senior's notion of 'economic man', Mill observes "political economy proceeds….under the supposition that man is a being who is determined, by the necessity of his nature, to prefer a greater proportion of wealth to a smaller in all cases…" Mill does consider "economic man" as fictional man to facilitate the propositions of political economy. The pages on 'economic man' in Mill's essay are followed immediately by the characterization of political economy as "essentially an abstract science; that employs "the method of a priori". The "disturbing causes" and "the tendency laws" in political economy necessitate *ceteris paribus* clauses.

4.2.4 J.E. Cairnes

J.E.Cairnes in his **Character and Logical Method of Political Economy (1875),** goes on to argue that political economy is a hypothetical, deductive science, and its conclusions "will correspond with facts only in the absence of disturbing causes, which is, in other words, to say that they represent not positive but hypothetical truths". By late 19th century there were growing doubts on the method of political economy. J.N.Keynes tried to argue that what is referred to as deductive reasoning in political economy was indeed preceded by inductive inferences based on observation of human behaviour.

4.2.5 J. N. Keynes

J.N. Keynes, in **The Scope and Method of Political Economy (1890),** argued that deductive reasoning was indeed preceded by concrete observations. Deductive reasoning comes at a later stage drawing upon specific individual experiences. J.N.Keynes is known for comprehensive reconciliation of the classical methodological approaches.

The following observation stands as a testimony to his comprehensive understanding of the methodological approaches underlying a complex subject like economics. He observes "…economic science deals with phenomena that are more complex and less uniform than those with which the natural sciences are concerned; and its conclusions, except in their most abstract form, lack both the certainty and the universality that pertain to physical laws. There is a corresponding difficulty in regard to the proper method of economic study; and the problem of defining preconditions and limits of the validity of economic reasonings become one of exceptional complexity. It is, moreover, impossible to establish the sight of any one method to hold the field to the exclusion of others. **Different methods are appropriate, according to the material available, the stage of investigation reached, and the object in view; and hence arises the special task of assigning to each its legitimate place and relative importance".** One could read this even to this day, as a manifesto of methodological pluralism, that is increasingly thought of as appropriate for economics. This aspect is further discussed, when we turn to Sen on economic methodology.

Check Your Progress 1

1) Explain the characteristics of Ricardo's method of explanation.

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2) State the general prepositions advanced by N.W.Senior in support of scientific status of the Ricardion system.

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3) What is methodological pluralism?

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4.3 ROBBINS, POSITIVISM AND APRIORISM IN ECONOMICS

Lional Robbins in his classic tract, *An Essay on the Nature and Significance of Economic Science (*1935 ed.) presented the dominant methodological view point of the early twentieth century economic thought, which stressed **subjectivism, methodological individualism, and the self-evident nature of the basic postulates of economic theory**. His objective was to counter the criticism from the historical and institutional schools that basic economic propositions were not amenable for Quantification, therefore not testable and hence not eligible to be called a science. Robbins countered by suggesting that basic propositions of Economics were deductions from a series of postulates. Chief of these postulates include the assumption involving the experience of the way in which scarcity of goods shows itself in the world of reality and how individuals order their preferences. These are universally acknowledged facts of experience. Robbins offers a coherent account of the status of terms used in economic theory. The fundamental assumptions are combined with subsidiary hypotheses which allow us to apply the theory to actual situations.

For Robbins rationality as a postulate in economics doesn't merely mean maximizing behaviour, but implies consistency of choice. Rationality as consistency of choice is stated in the sense that if A is preferred to B and B to C, A will be preferred to C. Robbins asserts that the assumption of rational conduct, and with it the assumption of perfect foresight in Economics, are 'expository devices'. They are not realistic assumptions. According to Robbins, empirical studies may be of use for the short-term prediction of possible trends, but they do not provide the grounds for discovering 'empirical laws'. The proper uses of realistic (empirical) studies are three: to check on the applicability of theoretical constructions to particular concrete situations, to suggest auxiliary postulates to be used with the fundamental generalizations, and to bring to light areas where pure theory can be reformulated or extended.

Robbin's insistence on Economics being based on true postulates born out of experience, which need not be tested, brings him closer to *apriorism* of the Austrian School. Economic laws are derived from postulates based on a *priori* truths of experience and therefore there is no need for testing them. Robbins' claim that Economics is a pure theory and still a Positive science may be somewhat puzzling. His claim of Positivism for Economics appears to be based more on his insistence that Economic laws are value- free. He argued that there was no room for ethical considerations in Economics, and in that sense too it is 'pure theory'.

Robbins' claim of 'pure theory' or apriorism status to Economics provoked devastating criticism, especially from T. Hutchison. The point of criticism is the 'emptiness of the propositions of pure theory'. If Economics is to be called as a positive science, Hutchison insisted that it should be testable. Otherwise it would end up as a '*pseudo-science'*. Second, the assumption of '*perfect expectations'* is baseless and therefore the rationality postulate is unrealistic. Third, there is need for more extensive use of empirical techniques in Economics instead of merely claiming all postulates are prior truths. Finally, Hutchison calls the use of 'psychological method' (or introspection) as ground for asserting the fundamental postulates.

4.4 HUTCHISON AND LOGICAL EMPIRICISM IN ECONOMICS

Terence Hutchison's *The Significance and Basic Postulates of Economic Theory* (1938) marks a clear turning point in the Twentieth Century Economic methodology and laid the foundation for a clear logical positivist turn to the practice of Economics. Hutchison, in his book, sets out on search for and making clear the foundations of modern economic theory. He states that economics is a science and as such it must appeal to facts, otherwise one would be engaging in 'pseudo-science'. What sets apart the empirical propositions of science from those of other intellectual endeavours is their testability.

According to Hutchison, science contains statements which are either conceivably falsifiable by empirical observation or are not. Those which are not falsifiable are tautologies and are thus devoid of empirical content. The primary reason why the propositions of pure theory have no empirical content is that they are posed in the form of deductive inferences. In Economics, widespread use of *ceteris paribus* clause robs empirical content. Irresponsible use of *ceteris paribus* clause make microeconomic theory effectively unfalsifiable.

Hutchison examines the formal structure of economic theory which consists of a series of deductions from basic postulates. These deductions are analytical in nature. These analytical statements are logical statements without any empirical content. Therefore, pure economic theory is empty and thus there is need for more empirical content. Hutchison insisted on the need for testability, and falsification tests even for the basic assumptions. The insistence of testing not only the theory, but the assumptions as well, for validity sparked-off a wider debate in economics later. But the immediate response was to describe Hutchison as an 'ultra-empiricist'.

Hutchison's attempt to establish the analyticity of fundamental generalizations of economic theory and to wean economists away from pure theory did not succeed very much. But his proposal to make economics increasingly testable or falsifiable form; to increase empirical investigations of various aspects of economics, and the need for economists to abandon their psychological method received universal acceptance by economists.

One of the interesting aspects of Hutchison's contribution relates to the issue of testing assumptions as well as the theory or hypothesis. This provoked a debate between Hutchison and Fritz Machlup. Machlup believes that only deduced or 'lower level' hypotheses require 'verification', and he outlines how such testing could be carried out. While Hutchison does not require that every statement in a theory be tested, he does insist that each be 'testable' and one should be able to conceive how a test could be carried out. Further, Hutchison prefers that the behavioural postulates of economics reflect the actual observed and statistically recorded behaviour of economic agents. Machlup requires no such correspondence. The crucial behavioural postulate is the assumption of rational, maximizing behaviour. And Machlup is more reasonable in arguing that this assumption need not be tested directly, which was insisted by Hutchison, but indirect testing of the same could be done. Much before the debate on testing of assumptions sparked-off by Hutchison's contribution, Milton Friedman's